Practical No 8
API Creation and Integration

A. Create a simple REST API using Express

Step 1: Create a project folder

Open VS Code → Terminal:

>mkdir mern-api
>cd mern-api

Step 2: Initialize Node.js
>npm.cmd init -y

Step 3: Install Express
>npm.cmd install express

Step 4: Create server.js

Code:

```
const  express  = require("express");
const app = express();
app.use(express.json());
app.listen(5000,()=>{
   console.log("server is running on 5000");
});
```

Step 5: Run the server
>node server.js

Output:

```
PS C:\Users\student\mern-api> node server.js
server is running on 5000
```

B. Implement a GET API to retrieve data.

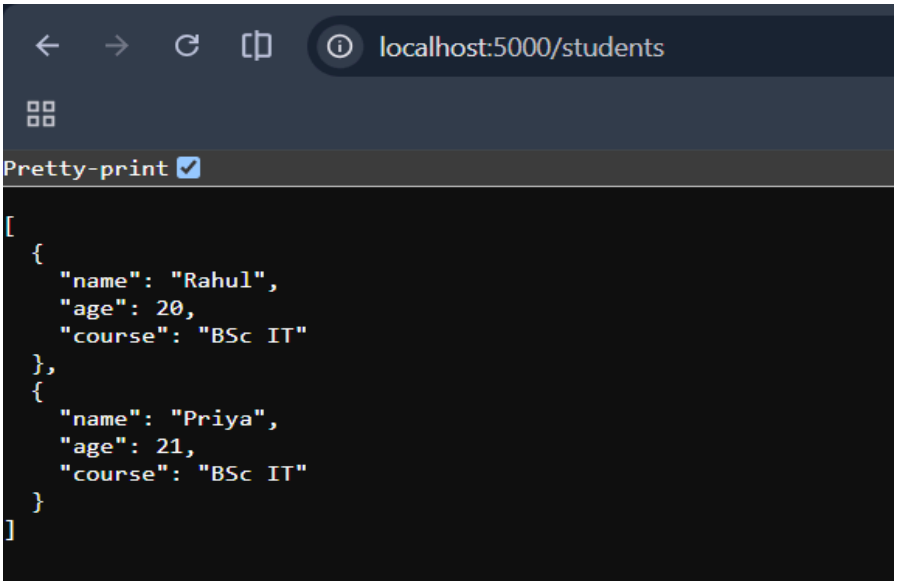
Code:

```
const express = require("express");
const app = express();
app.use(express.json());
app.get("/students", (req, res) => {
   res.json([
      { name: "Rahul", age: 20, course: "BSc IT" },
      { name: "Priya", age: 21, course: "BSc IT" }
   ]);
});
app.listen(5000, () => {
   console.log("Server running on port 5000");
});
```

Run:
>node server.js

Open browser:
http://localhost:5000/students

Output:



C. Fetch data from the API in a React application.

Step 1: Create React application

>npm create vite@latest student-frontend

Select:
React
JavaScript

Then:
>cd student-frontend

Step 2: Install Axios & CORS

In the React project:
>npm install axios

In the mern-api folder:
> npm.cmd install cors

Step 3:

App.jsx

```
import axios from "axios";
import { useEffect, useState } from "react";
function app() {
  const [students, setStudents] = useState([]);

  useEffect(() => {
    axios
      .get("http://localhost:5000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  },[]);

  return(
    <div>
      <h1>Student List</h1>
```

```
      {students.map((student,idx)=>(
        <p key={idx}>
          {student.name}-{student.age}-{student.course}
        </p>
      ))}
    </div>
  )
}
export default app;
```

server.js

```
const  express  = require("express");
const cors = require("cors");
const app = express();
app.use(cors());
app.use(express.json());

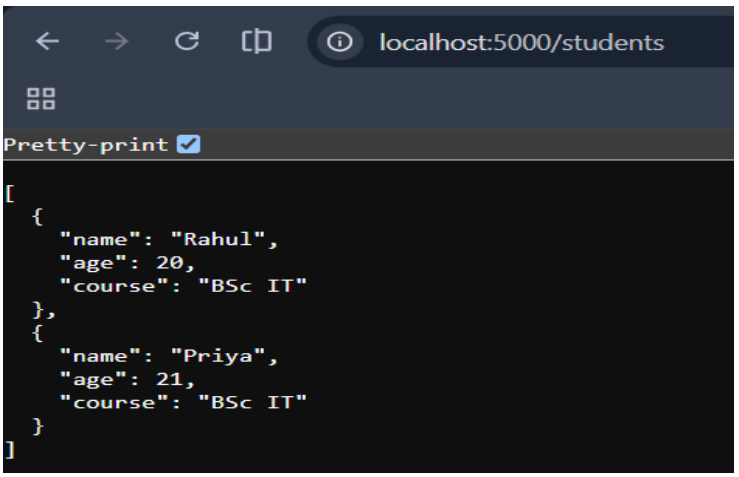
app.get("/students", (req, res) => {
    res.json([
        { name: "Rahul", age: 20, course: "BSc IT" },
        { name: "Priya", age: 21, course: "BSc IT" }
    ]);
});
app.listen(5000,()=>{
    console.log("server is running on 5000");
});
```

Run both projects

Terminal 1 — backend:
>cd mern-api
>node server.js

Output :
Server running on port 5000

Open the URL
http://localhost:5000/students



Terminal 2 — frontend:
>cd student-frontend
>npm run dev

Output:
Open the React URL shown by Vite, usually:

http://localhost:5173/



D. Send data from React to the server using a POST request.

Step 1: Add POST route in server.js

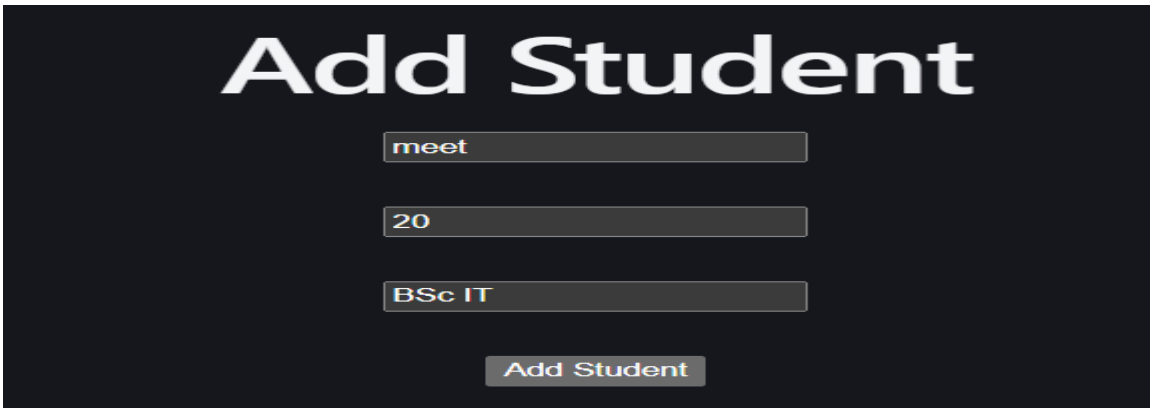
```
app.post("/students", (req, res) => {
    const student = req.body;
    console.log(student);
    res.json({
        message: "Student data received",
        student: student
    });
});
```

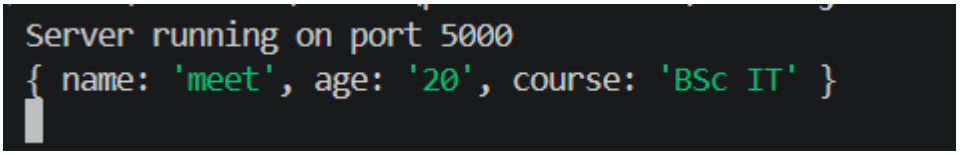
Step 2: Create a form in React

```
import { useState } from "react";
import axios from "axios";
function App() {
  const [name, setName] = useState("");
  const [age, setAge] = useState("");
  const [course, setCourse] = useState("");
  const addStudent = async (e) => {
    e.preventDefault();
    const student = {
      name: name,
      age: age,
      course: course,
    };
    const response = await axios.post(
      "http://localhost:5000/students",
      student,
    );
    console.log(response.data);
  };
  return (
    <div>
      <h1>Add Student</h1>
      <form onSubmit={addStudent}>
        <input
          type="text"
          placeholder="Enter name"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />
        <br />
        <br />
        <input
          type="number"
          placeholder="Enter age"
          value={age}
          onChange={(e) => setAge(e.target.value)}
        />
        <br />
        <br />
        <input
          type="text"
          placeholder="Enter course"
```

```
          value={course}
          onChange={(e) => setCourse(e.target.value)}
        />
        <br />
        <br />
        <button type="submit">Add Student</button>
      </form>
    </div>
  );
}
export default App;
```

Output:





E. Store data in MongoDB.

Step 1: Install Mongoose

In the backend folder:
>npm install mongoose

Step 2: Connect MongoDB

Step 3: Write Code in **sever.js**

```
const express = require("express");
const mongoose = require("mongoose");
const cors= require("cors");
const app = express();
```

```
app.use(cors());

// Middleware
app.use(express.json());


// Connect to MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/studentDB")
    .then(() => {
        console.log("MongoDB connected");
    })
    .catch((error) => {
        console.log("MongoDB connection error:", error);
    });


// Student Schema
const studentSchema = new mongoose.Schema({
    name: String,
    age: Number,
    course: String
});


// Student Model
const Student = mongoose.model("Student", studentSchema);


// POST API - Save student
app.post("/students", async (req, res) => {

    try {

        const student = new Student(req.body);

        await student.save();

        res.json({
            message: "Student saved successfully"
        });

    } catch (error) {

        res.status(500).json({
            message: "Error saving student",
            error: error.message
        });
```

```
    }

});


// Start Server
app.listen(3000, () => {
    console.log("Server running on http://localhost:3000");
});
```

Step 4: write code in App.js

```
import React, { useState } from "react";
import axios from "axios";

function App() {

    const [name, setName] = useState("");
    const [age, setAge] = useState("");
    const [course, setCourse] = useState("");

    const handleSubmit = async (e) => {

        e.preventDefault();

        try {

            const response = await axios.post(
                "http://localhost:3000/students",
                {
                    name: name,
                    age: Number(age),
                    course: course
                }
            );

            alert(response.data.message);

        } catch (error) {

            console.log(error);
            alert("Error saving student");

        }
    };
```

```
  return (
    <div>

      <h1>Student Form</h1>

      <form onSubmit={handleSubmit}>

        <input
          type="text"
          placeholder="Enter Name"
          value={name}
          onChange={(e) => setName(e.target.value)}
        />

        <br /><br />

        <input
          type="number"
          placeholder="Enter Age"
          value={age}
          onChange={(e) => setAge(e.target.value)}
        />

        <br /><br />

        <input
          type="text"
          placeholder="Enter Course"
          value={course}
          onChange={(e) => setCourse(e.target.value)}
        />

        <br /><br />

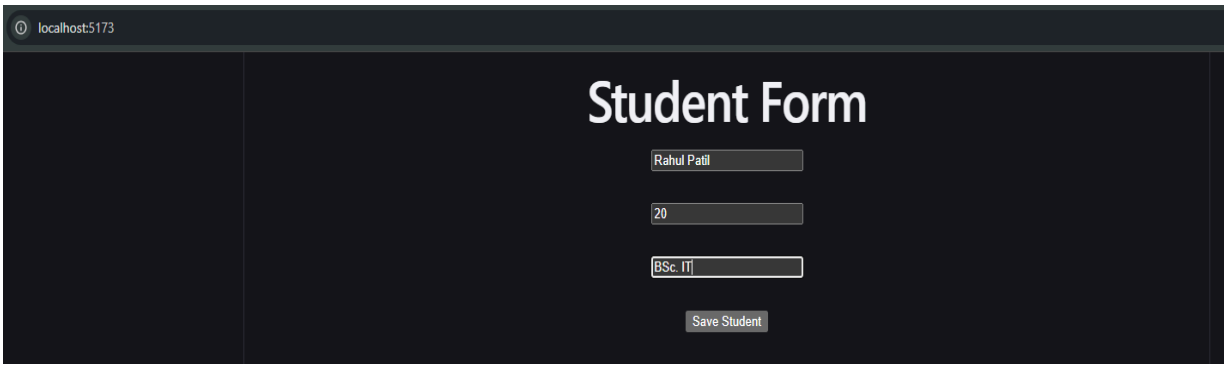
        <button type="submit">
          Save Student
        </button>

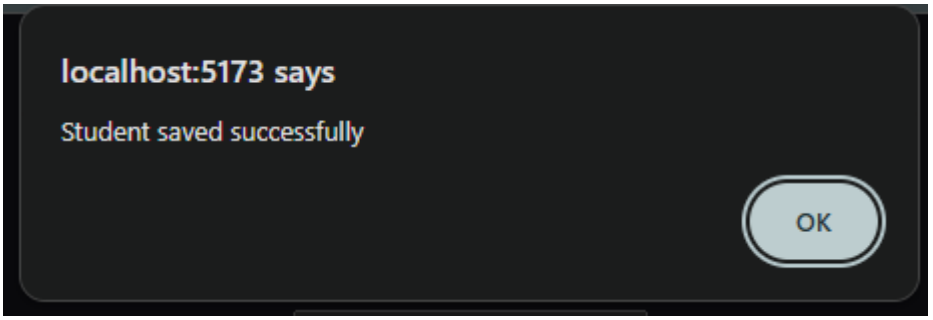
      </form>

    </div>
  );
}

export default App;
```

Output:


localhost:5173
Student Form
Rahul Patil
20
BSc. IT
Save Student


localhost:5173 says
Student saved successfully
OK

F. Display stored data in React page.

Step 1: Create GET API

In server.js:

```
app.get("/students", async (req, res) => {
 const students = await Student.find();
 res.json(students);
});
```

Step 2: React fetches the data

```
useEffect(() => {
    axios.get("http://localhost:5000/students")
          .then((response) => {
      setStudents(response.data);
    });
  }, []);
```

Step 3: Display it

```
{students.map((student) => (
       <div key={student._id}>
         <h3>{student.name}</h3>
         <p>Age: {student.age}</p>
         <p>Course: {student.course}</p>
       </div>
     ))}
```

Output:

